

Our vehicles: We are developing new standards for climate protection on the road

GRI 103-1

The Paris Agreement on climate protection establishes the goal of limiting global warming to significantly less than two degrees Celsius compared with the preindustrial level. In their climate protection plans, the signatory countries define their ambitious goals for reducing greenhouse gases overall and in individual sectors. For example, the climate protection plan of the German government aims to reduce total CO₂ emissions in the transport sector by 40 to 42 percent relative to 1990 levels by 2030.

EU regulations focus on new vehicles and prescribe the following reduction targets: By 2030, the CO₂ emissions of cars should be reduced by 37.5 percent relative to the base values of 2021; those of vans should be reduced by 31 percent. An intermediate target of 15 percent to be achieved by 2025 has been set for each of these two vehicle groups. Heavy-duty commercial vehicles must reduce their CO₂ emissions by 15 percent on average relative to the reference period 2019/2020 by 2025 and by 30 percent on average by 2030. Moreover, the overarching EU climate target within the framework of the EU Green Deal was adjusted from the current minus 40 percent to at least minus 55 percent for the period from 1990 to 2030. Against this background, we assume that the EU standards for new vehicles will also become much more stringent.

For us, the Paris Agreement represents more than just an obligation; our commitment to its targets stems from our fundamental convictions. We believe that it is our mission to develop technical innovations that will lead to CO₂-neutral mobility around the world. We realize that this mission will require a high level of investment. In order to finance it, we intend to increasingly use new tools such as [green bonds](#) in the future. Green bonds offer environment-oriented investors the opportunity to directly participate in the implementation of our technological strategy. However, the broad-based success of low-emission mobility requires not only sustainable investment but also favorable framework conditions. We need ambitious CO₂ pricing systems for fossil fuels and the creation of a comprehensive charging infrastructure as well as a hydrogen filling station network.

Our product development takes emissions into account from the start

GRI 103-2

Daimler has set itself the goal of developing products that are especially environmentally friendly and energy-efficient in their respective market segments. Our environmental and energy guidelines define how we intend to reach this goal. Product development plays a key role in this regard: A vehicle's

environmental impact — including its emissions of CO₂ and pollutants — is largely determined during the first phases of its development. The earlier in the development process we take environmental aspects into account, the more efficiently we can minimize the environmental impacts of our vehicles.

We systematically test the environmental friendliness of future products. An important tool in this process is the ongoing documentation of the development process. For every vehicle model and every engine variant, we define specific characteristics and target values — for example, for fuel consumption and pollutant emissions. We also use these target values to assess the milestones we pass in the course of product development. To this end we compare the current status of the project with the target values. If necessary, we initiate corrective measures on the basis of this assessment.

How we embed responsibility for more environmentally friendly vehicles in our organization

Our corporate management is responsible for setting our strategic goals, including targets for reducing our CO₂ emissions, and for monitoring our progress toward them. The Product Steering Board (PSB) monitors the development of the car fleet's CO₂ emissions in markets where such emissions are regulated. It is also responsible for providing forecasts. In its evaluations, the PSB takes into account the increasing degree of vehicle electrification and the changes that have been made to legal requirements, for example those related to the introduction of the new [WLTP](#) test procedure. The Board of Management then decides which measures need to be implemented. On the market side of the equation, price and volume control measures can also affect our ability to achieve our targets in the short term. For this reason, these measures are also discussed with the Board of Management within the framework of the regular reporting on the current state of [CO₂ fleet compliance](#).

Responsibility for ensuring compliance with climate protection requirements is split between several units and Board of Management members. The development units of the vehicle divisions are responsible at the vehicle level. For cars and vans, these are the "Drive Systems Product Group" development unit, the product groups of the vehicles, and Mercedes-Benz Vans Development; for trucks and buses, they are the "Global Powertrain & Manufacturing Engineering Trucks" unit and the vehicle divisions. The various directorates of the drivetrain development units also play a special role here. The heads of production are responsible at the level of the production plants, and the Heads of Sales at the Daimler showrooms.

Our CO₂ balance applies to the entire life cycles of our products

GRI 305-3

Most of our CO₂ emissions are generated during the use phase of the vehicles. But greenhouse gas emissions are also generated in other segments of a vehicle's life cycle, and we take that into account in our overall CO₂ balance sheet. We record the key figures we need for life cycle assessments and publish them in line with the basic principles of the [Greenhouse Gas Protocol](#).

In line with this leading global life cycle assessment standard, we divide our CO₂ emissions into three categories called the Greenhouse Gas Scopes. Scope 1 comprises all the emissions we cause ourselves through the combustion of energy carriers at our production locations, such as the generation of electricity

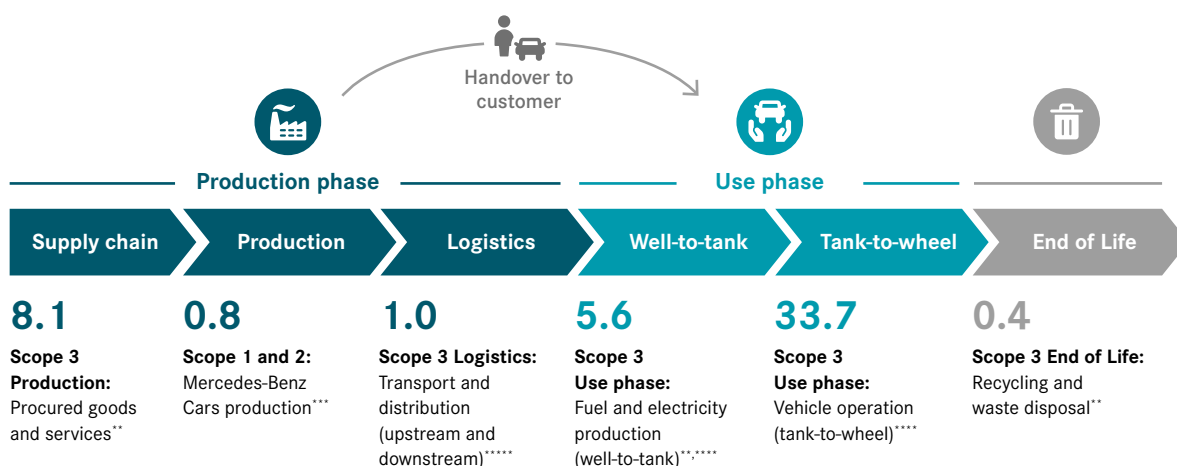
by our own power plants. Scope 2 includes all emissions that arise due to the generation of energy we purchase from external sources, such as electricity. Scope 3 includes all the emissions that are generated before (upstream of) or after (downstream of) our production operations. For example, Scope 3 includes the CO₂ emissions that arise in the supply chain (purchased goods and services), as a result of our vehicles' operation in customers' hands (the use phase, including the production of fuel and electricity) or in the recycling phase of the vehicles.

We have used these principles to calculate the emissions of the entire life cycle of the Mercedes-Benz car fleet worldwide. For 2020 we calculated an average CO₂ value of 49,7 tons per vehicle (see Diagram 9 in comparison with 2019 logistics were included for the first time).

Scope 3 emissions Mercedes-Benz Cars

9 | Scope 1, 2 and selected Scope 3 CO₂ emissions in tons per vehicle Mercedes-Benz Cars (2020)

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* For calculation basis see appendix [How we calculate and document our CO₂ emissions](#) and [Scope 3 emissions Mercedes-Benz Cars](#)

** See [life cycle assessment of vehicles](#)

*** See [key figures environment](#)

**** Driving emissions of Mercedes-Benz Cars fleet (EU, China, USA and RoW) standardized, mileage: 200,000 km, for data basis see chapter [Climate protection: Our CO₂ emissions – in all of our fleets](#)

***** Forecast value

Across all divisions: Our future is electric

We are developing all-electric and electrified model variants for all of our vehicle types – from cars and vans to trucks and buses. Thanks to our modular development approach, we can quickly transfer technologies between our divisions.

Our development focus is on battery-electric mobility for cars and vans, and on all-electric drive systems with batteries or fuel cells for trucks and buses.

That's why Daimler is also continuing to develop highly efficient combustion engines that can also be powered by [e-fuels](#).

That's yet another way we can help to further reduce the CO₂ emissions of future vehicle generations. However, e-fuels are still very expensive to produce today. In order to further develop them and bring them into circulation, we need suitable regulatory approaches. One important regulation in this area is the [Renewable Energy Directive](#) of the EU, which sets ambitious quotas for bringing synthetic fuels (including green hydrogen) into circulation. We are actively supporting this regulation through the political representation of our interests.

Partnerships

10 | Alternative drive systems Mercedes-Benz Cars*

		2019		2020	
Worldwide	Hybrid	29,907	1.2%	115,191	5.2%
	Electric drive	19,622	0.8%	47,672	2.2%
	Alternative drive systems (total)	49,529	2.0%	162,863	7.4%
MBC unit sales (total)		2,456,347		2,202,579	
Europe	Hybrid	16,091	1.7%	104,113	14.1%
	Electric drive	18,419	1.9%	42,711	5.8%
	Alternative drive systems (total)	34,510	3.6%	146,824	19.9%
MBC unit sales (total)		954,912		738,957	

* retail Mercedes-Benz Cars (incl. V- and X-Class)

As part of the effort to decarbonize transportation, Daimler Trucks & Buses relies on two complementary technologies: all-electric drive systems that use either batteries or fuel cells. By offering these two options, Daimler Trucks & Buses can optimally meet its customers' needs. The following rule of thumb applies: The lighter the payload and the shorter the distance, the more suitable a battery-electric truck is for the job. The heavier the payload and the longer the distance, the more a fuel-cell vehicle becomes the better option.

In November 2020 the Volvo Group and Daimler Truck AG signed a binding agreement to establish a joint venture for the development to series maturity of fuel-cell systems, their production, and their marketing. The focus is on utilization in heavy-duty trucks, but the systems will also be offered for other applications. By joining forces, the two companies can reduce their development costs and accelerate the market launch of the fuel-cell systems.

Mercedes-Benz Cars & Vans is pursuing ambitious climate-related goals

As part of our sustainable business strategy, Mercedes-Benz Cars & Vans has set itself the following goals:

- CO₂ neutrality for its new fleet by 2039: This applies to all the stages of automotive value creation – from the supply chain to production, the vehicle use phase, and vehicle disposal and recycling.
- Plug-in hybrids or all-electric vehicles will account for more than 50 percent of our car and van sales by 2030. This transformation will take place in a number of stages: Between now and 2022, we want to offer our customers a variety of electrified alternatives in every segment, ranging from smart cars to transporters. Depending on how circumstances develop, we at Mercedes-Benz Cars want the proportion of all-electric vehicles to increase to up to 25 percent of our total sales by 2025.

- At Mercedes-Benz AG we also plan to reduce the greenhouse gas emissions of the new vehicle fleet during the vehicle use phase ([well-to-wheel](#)) by more than 40 percent relative to 2018 by 2030. This target has been confirmed by the [Science Based Targets Initiative](#).

Reaching CO₂ neutrality as a team

At Mercedes-Benz AG, an interdisciplinary team consisting of environmental experts, buyers, developers, logistics specialists, production specialists, strategists, and sales experts monitor and manage CO₂ emissions in order to reach the goal of having a CO₂-neutral new-car fleet by 2039.

The Corporate Environmental Protection unit, for example, calculates the CO₂ emissions of all model series and all drive types at Mercedes-Benz Cars and conducts environmental and life cycle assessments for the vehicles. The Mercedes-Benz procurement unit works together with more than 2,000 direct suppliers so that it can also identify potential environmental risks in the supply chain and use targeted measures to minimize them. At the logistics unit, the routes for unavoidable goods transfers have to be optimized and the best mode of transport has to be used for each trip. That's the only way to reduce the emissions that are generated during goods delivery, sales operations, and shipment to distribution centers. Because most CO₂ emissions are generated during the use phase of a vehicle, CO₂ strategists are also important members of the team. These experts are responsible for the [tank-to-wheel](#) phase of a vehicle's life cycle, and through their know-how they help reduce the CO₂ emissions generated during driving operation. The working group also addresses the development of additional levers, such as CO₂-neutral production and a sustainable charging concept for Mercedes customers. Together, the members of the team are helping to ensure the successful decarbonization of Mercedes-Benz vehicles.

per charging point enables correspondingly designed vehicles to charge their batteries much faster than was previously possible. All of the IONITY charging points are integrated into the Mercedes me Charge system.

charge@home – simply charging at home

The new Mercedes-Benz Wallbox Home with a charging capacity of up to 22 kW and permanently attached charging cables enables users to charge their vehicles quickly for personal or private purposes using alternating current at home or at their workplace. This enables users to charge their battery-electric and plug-in hybrid vehicles conveniently. The charging capacity of 22 kW makes charging a vehicle much faster and more reliable than using a household socket would be.

In addition, the customers in the various markets can use smart wallboxes provided by national partners. Moreover, we offer our customers a comprehensive installation service for our range of wallboxes. This service includes on-site checks regarding the installation conditions, in-depth advice, and the installation itself, of course.

How we are promoting a charging infrastructure for our employees

In order to make electric mobility more appealing to our employees, we are continually expanding the charging infrastructure at our own locations. We have set up more than 3,500 charging points since 2013. At the end of 2020, our employees at Daimler locations in more than 50 German cities could take advantage of comprehensive charging solutions. In the charge@Daimler project, we are consolidating our efforts to establish an intelligent charging infrastructure at all Daimler locations in Germany. The project includes equipping parking lots, parking garages, and customer centers, as well as electrifying the in-house development test rigs and testing facilities. Within the company we have also ensured that our charging stations are provided with 100 percent certified green electricity. To make sure this is the case, we use certificates of origin that confirm climate-neutral power generation through the use of renewable energy sources.

We are forging ahead with the expansion of a hydrogen refueling station network

The network of hydrogen refueling stations is also growing. In the joint venture H2 MOBILITY Deutschland, we have been working together with Air Liquide, Linde, OMV, Shell, and Total since 2015 to expand the hydrogen refueling station infrastructure throughout Germany. At the end of 2020 there were 88 publicly accessible hydrogen refueling stations dispensing at a pressure of 700 bar in Germany; seven stations are under construction, and twelve more are in the planning or approval phase. This means that six new stations were added to the H2 MOBILITY network in the reporting year. The partners' long-term goal is to establish a network consisting of as many as 400 hydrogen refueling stations. A special focus will be on the expansion of an infrastructure for heavy-duty truck applications. We are conducting similar infrastructure projects together with additional partners in Europe, the United States, China, and Japan.

At the beginning of December 2020 we signed an agreement with Linde AG concerning the joint development of the next generation of liquid hydrogen refueling technology for fuel cell-powered trucks. Through this partnership we aim to make refueling with hydrogen as simple and practicable as possible.

At the end of the reporting year we also joined forces with the Volvo Group, IVECO, OMV, and Shell in the H2 Accelerate (H2A) initiative. The aim of this initiative is to continue expanding the series production of hydrogen-powered trucks and a Europe-wide CO₂-free hydrogen refueling network.

How we assess the effectiveness of our management approach

GRI 103-3

In our management approach to climate protection, we are pursuing the goals of our "Ambition 2039" in order to live up to our responsibility as signatories to the Paris Agreement on climate protection. We have defined the measures we need to take in order to reach these goals. We use internal and external performance reviews to evaluate how well we are implementing these measures. Internally, the specialist units check their progress several times a year. Externally, we commission an auditing company to audit selected goals and implementation measures. In addition, Mercedes-Benz AG has defined a specific CO₂ reduction pathway in line with the standards of the Science Based Targets Initiative (SBTI). The SBTI has confirmed that this pathway conforms to the Paris Agreement on climate protection.

Furthermore, we conduct dialogs regarding climate protection and we use the knowledge gained to review our management approach and adjust it as needed. For example, we hold in-depth discussions with environmental institutes and NGOs during our annual ["Daimler Sustainability Dialogue"](#). We also conduct talks on the subject of climate protection with our Board of Management throughout the year. In addition, the feedback we continually receive from the government, the public, and our other stakeholders lets us know how the sustainability goals we have set for ourselves are being perceived and evaluated.

For many years now, reaching the CO₂ targets we have set for our fleets has been a criterion for determining the remuneration of our Board of Management. During the reporting year, we continued to differentiate this incentive system and expanded it to cover the entire senior management structure, from the department heads to the very top. We expect this process to motivate the entire Group even more strongly to reach the sustainability goals we have set for ourselves.

On the road to CO₂-neutral production

GRI 103-1

As part of our sustainable business strategy, we have set ourselves the overall goal of making the mobility of the future more sustainable. One important target is the reduction of greenhouse gas emissions. This applies not only to our mobility solutions but also to our own production plants. By pursuing our goal of making our production processes CO₂-neutral over the long term we are fulfilling our voluntary commitment to the Paris Agreement and complying with other national and international climate protection guidelines.

How we make our production more environmentally and climate-friendly

GRI 103-2

Daimler operates 72 production facilities all over the world that are subject to a variety of regional and national laws.

The environmental and climate protection measures at our production locations are controlled and coordinated across business units by three regional committees (Germany/Europe, North and South America, and Asia). These committees enable our experts in these theme fields to create networks linking companies and plants and serve to develop globally accepted standards and procedures. Their mission is to continuously improve our environmental and climate protection performance by developing standards, sharing tried-and-tested as well as innovative processes, and communicating our environmental goals.

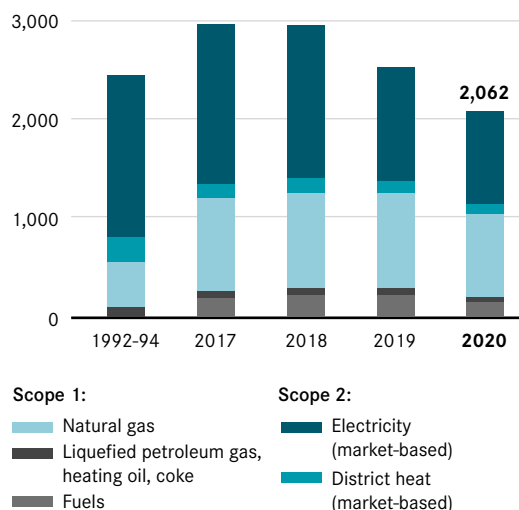
Climate protection goals for our plants

We have also set ambitious climate protection goals for our plants.

- **Starting in 2022:** CO₂-neutral production at our own Mercedes-Benz Cars and Vans production plants worldwide and at the Daimler Trucks & Buses production plants in Europe.
- **By 2025:** CO₂-neutral production at all Daimler Trucks North America (DTNA) production plants. The Portland truck manufacturing plant already achieved CO₂-neutral production in 2020.
- **By 2030:** At Mercedes-Benz Cars & Vans we are striving to reduce the CO₂ emissions at our plants (Scope 1 and 2) by 50 percent relative to the reference year 2018 by 2030. This target has been confirmed by the [Science Based Targets Initiative](#).
- **By 2039:** CO₂-neutral production at all of our plants and in all of our business units worldwide.

15 | Direct and indirect CO₂ emissions from production

in 1,000 t

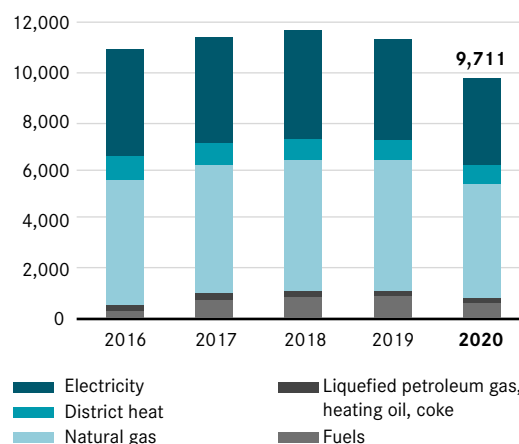


Our goal for the future is CO₂-neutral production

Mercedes-Benz AG is pursuing an ambitious target for its own production plants and the procurement of the energy they consume: Starting in 2022, the operations of Mercedes-Benz AG's own production locations — more than 30 car and van plants around the world — are to be CO₂-neutral. All of the company's bus and truck production locations in Europe are also planned to have CO₂-neutral production operations by the end of 2022 at the latest.

16 | Energy consumption in production

in GWh



We procure green electricity

In the first step toward CO₂-neutral production, we aim to reduce or — wherever possible — completely eliminate the greenhouse gas emissions that are generated by the plants' vehicle production and energy supply. One important pillar of this project is the establishment of a sustainable energy supply for the plants, for example by purchasing green electricity. Starting in 2022, all the locations of Daimler AG, Mercedes-Benz AG, Daimler Trucks AG, and Daimler Mobility AG in Germany plan to purchase 100 percent of their electricity from renewable energy sources to ensure that it is CO₂-free. An electricity contract for all Daimler locations will ensure that their energy needs will be fully covered at all times by a mix of wind, solar, and hydro-electric power. A large proportion of the green electricity that is purchased must be produced in Germany, for example in wind power stations whose subsidies have expired after 20 years of operation in line with the Renewable Energy Sources Act (EEG). One of the reasons for this stipulation is that it will safeguard the continued operation of six wind farms in northern Germany over the long term.

In the reporting period, many of our production plants were supplied with electricity from renewable sources. Examples include the production of the EQC electric vehicle (combined electrical consumption: 21.5–20.1 kWh/100 km; combined CO₂ emissions: 0 g/km)^{1,2} at the Mercedes-Benz location in Bremen and battery production at the Deutsche Accumotive GmbH & Co. KG plant in Kamen, Saxony. The van production plant in Vitoria (Spain), which produces the eVito electric models, also used only green electricity throughout the entire location. Since 2020 it has also manufactured the EQV (combined electrical consumption: 26.4–26.3 kWh/100 km; combined CO₂ emissions: 0 g/km)^{1,2}.

We generate green energy at more and more locations

GRI 302-1

Another major pillar of CO₂-neutral production involves increasing the production of energy from renewable sources at the various locations.

At the Sindelfingen location, Factory 56, Mercedes-Benz' most modern production facility, was commissioned during the reporting year. It too has been supplied with CO₂-neutral energy from the very start. A photovoltaic system installed on the roof of the production hall supplies green electricity for the manufacturing operations below. Factory 56 is also equipped with many sustainability features that reduce its energy consumption. About 40 percent of the roof's surface is covered with plantings, which offset the paved areas at the facility and also retain rainwater.

Factory 56 is supplied with direct current produced by the facility itself. This is the first Mercedes-Benz plant to install such a system; this is a technological milestone for the company. The photovoltaic system on the roof generates direct current and has a total output of more than 5,000 kWp. The power it generates is fed directly into the plant's own direct-current network without any inverters or power losses. As a result, it does

not need to be converted into alternating current, which would result in an energy loss of 10 to 15 percent. Fluctuations in the energy supply due to changing weather conditions or times of day are balanced out by a battery storage unit developed by Mercedes-Benz Energy GmbH with a total capacity of 1.4 MWh. In addition to making a measurable contribution to reducing energy consumption and CO₂ emissions, the new technology improves the supply security of the production operations and also stabilizes the local power grid, for example by smoothing the **peak loads**.

Many photovoltaic systems have also been installed at the production locations of Daimler Truck AG worldwide. One example particularly worth mentioning is the plant in Chennai, India, which covered about 16 percent of its energy needs in 2020 via its own photovoltaic system. Photovoltaic systems have also been commissioned in Japan (Kawasaki), the United States (Portland, Cleveland, High Point), Turkey (Istanbul), and Germany (Neu-Ulm, Rastatt, Gaggenau, and Kassel, for example). We are now surveying potential locations worldwide for the installation of additional photovoltaic systems. The continued expansion of our locations' on-site generation of green power is one of the pillars of decarbonization.

In the reporting year, renewable energy accounted for 60 percent of the total electricity consumption at Mercedes-Benz Cars production plants and for 26 percent of the total energy consumption. At Vans, it accounted for 62 percent of the total electricity consumption and 20 percent of the total energy consumption. The Trucks & Buses division achieved a proportion of 45 percent for the total electricity consumption and 18 percent for the total energy consumption.

The heat supply is becoming more sustainable

Mercedes-Benz AG is also using similar levers to boost the sustainability of the plants' heat supply. Among other things, the company is planning to use biogas and biomass as well as geothermal and solar thermal energy. It is also commissioning heat pumps powered by green electricity. At the Bremen plant, for example, CO₂ emissions are being significantly reduced, partly by the use of district heating. During the reporting year the Mercedes-Benz plant in Jawor (Poland) used such technologies for CO₂-neutral engine production and its production of plug-in hybrid batteries, which began this year. The plant will also derive electricity from a wind farm and cover its heat requirements by means of a biomass-pellet heating plant and a boiler that burns biomethane from renewable raw materials. Mercedes-Benz AG is thus one of the first large industrial companies in Poland to have signed contracts with local green power and heat supply utilities. We are currently looking at additional plants and measures.

Unavoidable emissions are offset

To supplement these measures, starting in 2022

Mercedes-Benz AG aims to implement qualified climate protection projects to offset all the CO₂ emissions that so far have been impossible to avoid. In a first step, Daimler Truck AG is taking the same approach for Europe. Unavoidable emissions

1 Electricity consumption and range were calculated on the basis of Commission Regulation (EC) No. 692/2008. Electricity consumption and range depend on the vehicle configuration.

2 The actual range is also dependent on individual driving style, road and traffic conditions, outside temperature, use of air conditioning/heating systems etc. and may therefore differ.

We have also prepared ourselves for the fuel emissions trade in Germany

Since the beginning of 2021, Germany has also had a legally prescribed fuel emissions trading process that complements the European emissions trading scheme. The new Fuel Emissions Trading Act (BEHG) has introduced CO₂ pricing as part of a national emissions trading process for amounts that are not already subject to the EU Emissions Trading Scheme (EU ETS). The law applies to the heating and transportation sectors in particular. This means that we must acquire certificates for the fossil energy carriers we use that are not subject to the EU ETS.

In order to adapt to this new legislation, Daimler informed the locations and the responsible in-house managers about the legal requirements and the new tasks connected with them. In order to avoid double burdens arising from the European and the national emissions trade, we need to prove that a plant is already subject to the EU ETS, for example. Together with our fuel suppliers, we have therefore adapted our invoicing procedures and installed meters to provide the corresponding verification if it is needed. We have also implemented the corresponding processes for trading these certificates at Daimler.

Transport logistics are helping to protect the climate

Our global transport logistics operations currently serve 75 manufacturing plants in around 30 countries and about 8,500 retailers in almost every part of the world. We transported around 2.7 million vehicles worldwide in 2020. In addition, almost 4.5 million tons of production materials were transported within Europe in the first half of 2020 alone. Our global transport volume amounted to around 350,000 standard containers of sea freight and about 120,000 tons of air freight. The measures taken against the global covid-19 pandemic caused a 15 percent decrease in this volume relative to 2019.

We are working hard to optimize our logistics network in order to reduce the associated CO₂ emissions. Our main goal is to optimally connect transportation hubs with one another so as to reduce the distances traveled and utilize capacity more efficiently. Innovative transportation concepts and new modes of transport also play a major role here.

We select logistics concepts not only on the basis of their costs, duration, and transport quality, but also according to their CO₂ emissions. When selecting providers of logistics services, we also take sustainability criteria into account – ranging from environmental certificates and the use of environmentally compatible equipment to the utilization of low-emission trucks that meet the latest Euro emissions standards.

Mercedes-Benz Cars is continually working to optimize its global logistics network. In particular, the company is steadily increasing the volumes it transports via the railroad network and focusing on sustainable and digitalized innovations. In its implementation efforts, Mercedes-Benz Cars is now taking the next step toward CO₂-neutral mobility: At the beginning of 2020,

Mercedes-Benz cooperated with its logistics partner DB Cargo to switch the transport logistics to a CO₂-free energy supply. From now on, the production material for the Mercedes-Benz car plants in Germany and the plant in Kecskemét (Hungary) will be transported in trains powered by green electricity. The volume of these rail shipments is the equivalent of about 270 truckloads per day, which do not need to be transported by road.

The green electricity that is used for rail shipments in Germany and Austria is supplied exclusively by local sources of renewable energy. Today it is mainly generated by hydroelectric power plants.

In addition to the focus on reducing greenhouse gas emissions, activities to reduce traffic volume, waste generation, and energy demand were also pursued.

In this connection we also launched a pilot project for reducing the greenhouse gases generated by local transportation at the production locations. The first plant to participate was that in Kecskemét. Starting in May 2020, a fleet of seven Mercedes-Benz Actros trucks has been using only an alternative fuel (HVO diesel) as they complete their seven daily rounds of up to 100 deliveries. As a result, the plant was already able to reduce its CO₂ footprint by at least 58 tons (this corresponds to approximately the amount of emissions that would have been generated by a megatrailer driving 15 times the distance between Madrid and Moscow using conventional diesel fuel).

The aim of this pilot project is to let the findings from the test operations flow into the CO₂ reduction strategy of other plants.

How we assess the effectiveness of our management approach

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On the road to CO₂-neutral production, we have already achieved success in a number of areas. In 2019 we already reached the long-term CO₂ reduction targets we had wanted our production operations to reach by 2020. In 2020 Mercedes-Benz AG reduced its CO₂ emissions by 18 percent relative to the previous year partly thanks to a variety of efficiency-boosting measures to decrease greenhouse gas emissions in production processes. These are systematically documented in a Group database that is used to control progress toward Group-wide goals. Greenhouse gas emissions were also reduced by 20 percent at Trucks & Buses.

Mercedes-Benz AG and Daimler Truck AG have set new future goals for themselves as part of their “Ambition 2039” and as an expression of their commitment to the Paris Agreement on climate protection.

[Climate protection goals for our plants](#)

We use internal and external tools to monitor the progress we are making toward the climate protection goals we have set for our production plants. We use the results of our reviews to adapt and refine our climate protection measures. Mercedes-Benz AG and Daimler Truck AG have defined the

parameters for in-house reviews, and they monitor these parameters by means of a scorecard. We have commissioned an auditing company to conduct the external review. This company annually evaluates a selected number of our corporate goals and their implementation.

17 | CO₂ emissions from energy consumption (in 1,000 t)*

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	2016	2017	2018	2019	2020
CO ₂ direct (Scope 1)	1,056	1,192	1,247	1,239	1,027
CO ₂ indirect (Scope 2) - market-based	1,882	1,763	1,687	1,276	1,035
CO ₂ indirect (Scope 2) - location-based	2,141	2,041	1,985	1,706	1,492
Total - market-based	2,938	2,955	2,934	2,516	2,062
Total - location-based	3,197	3,233	3,232	2,946	2,519

* Since 2016, the "market-based" and "location-based" accounting approaches have been implemented in accordance with GHG Protocol Scope 2 Guidance. Since then, the market-based approach has been the standard accounting approach. The historical data for 2006-2015 were calculated using a method similar to the location-based approach.

18 | Specific CO₂ emissions (in kg/vehicle)*

GRI 305-1/-2

	2016	2017	2018	2019	2020
Cars – CO ₂ direct (Scope 1)	245	250	267	279	326
Cars – CO ₂ indirect (Scope 2) – market-based**	611	565	562	431	426
Total – Cars – Scope 1 & 2	856	815	829	711	752
Trucks*** – CO ₂ direct (Scope 1)	746	663	629	676	742
Trucks*** – CO ₂ indirect (Scope 2) – market-based**	1,286	1,084	933	834	954
Total – Trucks – Scope 1 & 2	2,032	1,747	1,561	1,510	1,696
Vans – CO ₂ direct (Scope 1)	372	340	355	346	333
Vans – CO ₂ indirect (Scope 2) – market-based**	201	157	196	160	147
Total – Vans – Scope 1 & 2	573	497	551	506	479
Buses – CO ₂ direct (Scope 1)	1,408	1,177	977	1,083	1,471
Buses – CO ₂ indirect (Scope 2) – market-based**	1,421	1,059	948	911	1,245
Total – Buses – Scope 1 & 2	2,829	2,236	1,924	1,994	2,716

* Excluding CO₂ from liquid fuels

** Since 2016, the "market-based" and "location-based" accounting approaches have been implemented in accordance with GHG Protocol Scope 2 Guidance.

*** Reman scopes have no longer been taken into account in the Trucks division since 2020.

Air quality: Better technologies comply with more stringent limits

GRI 103-1

Our corporate responsibility as an automaker includes our efforts to bring individual mobility, climate protection, and air quality into harmony. This is because road traffic still accounts for a considerable share of air pollution, especially in the vicinity of streets and roads. As a result, the air quality in cities is an important focus of our sense of responsibility for the environment. Legislators all over the world have set standards for emissions in order to regulate the emission of harmful substances such as nitrogen oxides (NO_x) and particulates and to reduce air pollution. These emission limit values have become ever more stringent over the past few years. In order to remain below these limit values today and in the future, we are continuously developing our technologies.

In addition to vehicle emissions, we are also paying attention to the airborne emissions of our production plants. Lowering the airborne emissions from our plants is a constant task and a challenge — for our plant and facility planning teams and our daily operations. Volatile organic compounds (VOCs) are especially important in this regard, in particular those produced in our paint shops. Other significant air pollutants include the nitrogen oxide and sulfur oxide emissions from our furnaces and energy generation systems, as well as particulate matter released by the welding smoke exhaust units in our bodysheet areas and by our energy generation systems.

How we plan to improve the air quality in cities

GRI 103-2

Plans call for our new vehicle fleet to no longer have any relevant impact on NO₂ emissions in urban areas by 2025. Another one of our aims is to forge ahead with the development of new measures for reducing particulate emissions.

New diesel engines comply with stricter limit values

A reduction of NO_x emissions is made possible by an innovative overall package consisting of the engine and the exhaust gas aftertreatment system. This package is being continuously enhanced and has been systematically launched on the market in the new engine generation encompassing the OM 654 and 656. The company has invested approximately €3 billion in development and production for this purpose. Vehicles equipped with the new engines also generate low NO_x emissions in real driving operation. On many journeys using the [Real Driving Emissions \(RDE\)](#) measuring process, they actually record values that are significantly lower than the current laboratory threshold limit of 80 milligrams per kilometer. Vehicles equipped with the latest generation of diesel engines achieve average NO_x values of around 20 to 30 milligrams per kilometer in long-term operation over many thousands of kilometers under RDE conditions.

Updating the fleet with new diesel vehicles certified according to the [Euro 6d-TEMP](#) or Euro 6d standards is a very effective measure to further reduce NO_x emissions in road traffic. At Mercedes-Benz Cars, the entire new car fleet for Europe has been certified according to the Euro 6d-TEMP standard or better since June 2019 and according to Euro 6d since the fall of 2020. This was made possible by the expanded exhaust gas aftertreatment system using an additional [underfloor SCR catalytic converter](#), as well as other measures.

We are implementing a plan of concrete measures

In addition to the previously mentioned introduction of vehicles that fulfill the more stringent emissions limit values as part of the RDE requirements, we are implementing a series of additional measures to further improve air quality.

A software update that reduces nitrogen oxide emissions

Overall, Daimler is developing software updates for almost the entire fleet of Euro 6b and Euro 5 diesel vehicles in Europe. These updates reduce the nitrogen oxide emissions of the vehicles in real driving operation by 25 to 30 percent on average.

As early as 2017, Daimler announced that it would offer voluntary service measures that include software updates for several million diesel vehicles in Europe. Daimler has also been carrying out obligatory recalls — during which software updates are also applied — at the order of Germany's Federal Motor Transport Authority (KBA) since 2018.

The recalls at the order of the KBA apply to a number of vehicle models (cars and vans) that comply with the Euro 6b or Euro 5 emission standards. The voluntary service measure for vehicles that are not included in the recall is continuing as planned.

We are promoting hardware retrofitting in priority regions

In the previously defined priority regions, we are participating in a hardware retrofit program for diesel vehicles that was initiated by the German federal government. Specifically, Daimler has agreed to provide a financial subsidy of up to €3,000 (gross) per vehicle for hardware retrofitting if certain conditions are met. The hardware retrofitting must be developed and offered by a third-party supplier and approved by the German Federal Motor Transport Authority (KBA). In the summer of 2019, the KBA approved retrofitting solutions for various vehicle models. The retrofitted vehicles must comply with the NO_x limit value of 270 mg/km in real driving operation under specific conditions. The aim is to guarantee a significant decrease of the retrofitted vehicles' NO_x emissions in permanent operation.

To make it as easy and efficient as possible for our customers to apply for the Daimler grant, we have set up a [special website](#) for this purpose. Interested parties visit this website in order to find out whether they fulfill the precise requirements for receiving the grant. Customers can also use this website to request payment of the grant after the approved retrofit hardware has been installed.